

eSports Tournament Management System

Group Number: _____ **Course:** Database Systems

Group Members: Add name and roll number of each member here.

Project Summary

The eSports Tournament Management System is a relational database project designed to manage competitive gaming tournaments. The system stores and organizes data about players, teams, games, tournaments, registrations, matches, scores, venues, and sponsors. It supports real-world tournament operations such as registering players, creating teams, registering teams into tournaments, scheduling matches, recording results, and tracking sponsors.

Tools and Technologies Used

Tool	Purpose
MySQL	Relational database management system
MySQL Workbench	Database design and SQL query execution
Python Flask	Backend and database connectivity
HTML/CSS	Frontend user interface
GitHub	Version control and project repository
draw.io / diagrams.net	Crow's Foot ERD design

Database Concepts Demonstrated

Primary keys, foreign keys, composite keys, constraints, many-to-many relationships, associative tables, normalization up to 3NF, SQL queries, views, and relational schema design.

Main Entities

Teams, Players, TeamMembers, Games, Venues, Tournaments, Registrations, Sponsors, TournamentSponsors, Matches, and MatchScores.

Entity Relationship Diagram (ERD)

Crow's Foot style relationships are represented through one-to-many labels and associative tables for many-to-many relationships.

